

Bridging Visual Rhetoric and Computer Vision: A Large-Scale Analysis of the Design of Japanese and Original-Language Film Posters

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Abstract

This study connects visual rhetoric with computational and quantitative methods to reveal the cultural norms embedded in film poster design across languages and regions. We constructed a large-scale dataset of over 100,000 posters from The Movie Database (TMDB), covering foreign films released between 1960 and 2024 that also had Japanese versions. Using image processing techniques, we extracted visual features such as colour, text, faces, and objects, and conducted cross-linguistic comparisons. The analysis revealed that Japanese posters display substantially more visual information than their original counterparts. By introducing quantitative and statistical perspectives to a field that has largely relied on case studies, this research provides new academic contributions.

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1 Introduction

Since their emergence, film posters have functioned not only as advertising media but also as powerful cultural devices that shape audience desires, communicate values, and convey social and political messages [Staiger 1990]. From wartime propaganda to the Hollywood star system and contemporary global marketing, film posters have played a significant role in the formation of cultural norms [Smith 2018]. Despite this cultural significance, comprehensive research on film poster representation remains scarce both in Japan and abroad. Existing studies are mostly case-based, addressing only a few works, and aside from a limited number of overseas examples [Kagan et al. 2024; Song et al. 2024], no large-scale representational studies of film posters have been conducted.

Building on this context, the present study applies the framework of visual rhetoric, which views visual artefacts as persuasive texts using symbolic strategies to shape interpretation [Danesi 2017; Foss 2004]. While rhetorical and semiotic analyses have provided valuable insights into how posters construct meaning, these approaches

have generally relied on close readings of limited samples, making it difficult to capture broader patterns across cultures or time. To address this gap, we adopt a computational approach, applying large-scale analysis to original and Japanese-language film posters. By operationalising rhetorical features—such as text prominence, facial emphasis, and compositional density—through computer vision methods, we aim to illuminate systematic differences in design strategies across cultural contexts. In doing so, this research not only advances poster studies beyond case-specific interpretation but also contributes to broader discussions of how visual media shape and reflect cultural norms.

2 Methods

This research adopts Distant Viewing as its methodological framework for analysing large-scale visual data [Arnold and Tilton 2023]. Distant Viewing consists of four stages: (1) annotate, (2) organise, (3) explore, and (4) communicate. The analyses performed in this study—face detection, text detection, object detection, and main colour extraction—fall under the annotate stage. We also used a Personality Scoring Network (PSN) [Zhao et al. 2018] to estimate impression scores.

While not strictly an annotation of visual features, the PSN attempts to quantify the subjective atmosphere evoked by each poster, providing interpretive support for visual rhetoric research. The dataset was compiled from The Movie Database and consists of 100,202 posters for foreign films released between 1960 and 2024, including both Japanese and original-language versions. In this study, original-language posters refer to the posters whose language tag matches the `original_language` field in the film's metadata on TMDB, and which are therefore considered the design intended for the film's primary market of release. RetinaFace was used for face detection [Deng et al. 2020], CRAFT for text detection [Baek et al. 2019], YOLOv8x-seg for object detection [Jocher et al. 2023], and the Personality Scoring Network (PSN) for impression scoring. Colour analysis was performed with a custom program that determined cluster numbers using the Elbow method, extracted dominant colours, and measured colour concentration via the Herfindahl–Hirschman Index (HHI).

The extracted features were analysed with statistical tests such as Mann-Whitney U and χ^2 tests, enabling systematic comparison between Japanese and original posters. These techniques translate rhetorical features into measurable variables, enabling a systematic cross-cultural comparison.

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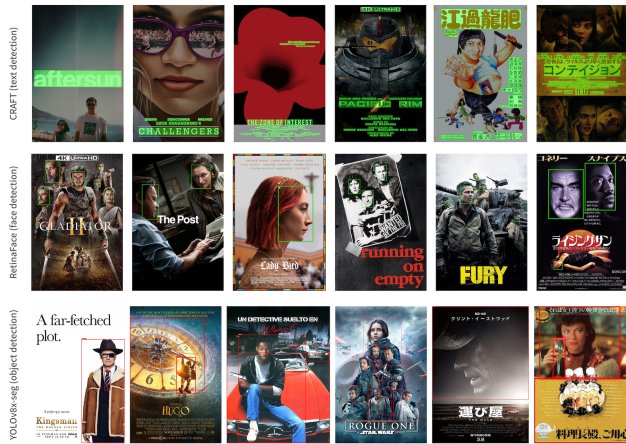


Figure 1: Examples of text detection using CRAFT (top), face detection using RetinaFace (middle), and object detection using YOLOv8x-seg (bottom).

3 Results

The analysis revealed significant differences between Japanese-language and original-language posters. Japanese posters contained more detected text regions (Mann–Whitney $U = 5.77 \times 10^8$, $p < 10^{-308}$), larger facial area ratios ($U = 4.41 \times 10^8$), and a greater number of depicted objects ($U = 4.10 \times 10^8$, $p = 1.48 \times 10^{-9}$). Colour diversity was also significantly higher, as indicated by lower HHI values ($U = 3.51 \times 10^8$, $p = 8.13 \times 10^{-73}$). Conversely, "minimalist" scores by PSN were significantly higher for original-language posters. Collectively, these findings demonstrate that Japanese-language posters, compared with their original-language counterparts, exhibit a systematic increase in information density. This pattern aligns with non-academic discourses, including online communities, such as websites and social media threads, that have long compared Japanese and original-language posters, often describing the former as "dense" or "overloaded" with taglines and faces. Crucially, however, our results provide the first large-scale, quantitative evidence supporting these claims.

On the basis of these findings, we argue that the information richness of Japanese posters is not incidental but reflects a culturally embedded rhetorical strategy of persuasion: "to present as much information as possible." This rhetorical technique extends beyond film posters, resonating with other aspects of Japanese design culture, such as food menus illustrated with photographs or plastic samples, as well as information-rich web user interfaces [Doi and Murata 2021]. In addition, considering the specific conditions of the Japanese film industry further enriches this analysis. Foreign films are often released later in Japan and promoted for shorter periods, requiring posters to convey key information at a glance. Moreover, as cinema attendance declines and general audiences become less familiar with foreign actors, directors, and franchises, film posters tend to compensate by foregrounding recognisable faces, award histories, and descriptive taglines to reduce uncertainty and attract viewers. As a result, these designs prioritise informativeness over minimalism.

4 Conclusion and Future Work

This study demonstrates that Japanese film posters consistently exhibit higher information density than their original counterparts and that this density functions as a rhetorical strategy of persuasion in Japanese design culture. By combining large-scale datasets with computational techniques, this research expands the possibilities of visual culture and visual rhetoric studies, which have previously relied heavily on case-based approaches.

This study also shows how cultural norms shape visual design, using film posters as a case study. By making visible the rhetorical conventions embedded in Japanese posters, the research contributes to broader debates on cultural translation and adaptation in visual media.

Future work will expand this framework by incorporating deep learning and clustering methods for a more nuanced analysis of visual representation.

Beyond this, we aim to extend the implications of this research to contemporary issues in AI-generated imagery.

Generative models are increasingly used to produce visual content for global audiences, but they often rely on training datasets that are predominantly shaped by Western cultural norms. As a result, cultural conventions from other regions—such as Japanese rhetorical preference for information-rich visual communication identified in this study—are underrepresented or misrepresented. Building on our findings, we plan to explore how culturally specific visual strategies can be formally modelled and integrated into generative systems, contributing to more culturally inclusive and context-aware image generation.

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